

Bureau of Indian Standards

Manak Bhawan, 9, Bahadur Shah Zafar Marg
New Delhi - 110002

IS 1758:2016

2016

TEXTILE FABRICS - DETERMINATION OF DIMENSIONAL CHANGES

SPECIFICATION

ICS 59.080.30

BIS 2016

Indian Standard

1 SCOPE

This standard specifies the method for determination of dimensional changes (shrinkage or extension) of woven and knitted textile fabrics when subjected to washing, drying, or ironing under specified conditions.

2 REFERENCES

- IS 1758 (Part 2) - Dimensional Changes after Home Laundering
- IS 2454 - Sampling Procedures for Textile Fabrics
- IS 3820 - Textiles: Standard Atmosphere for Conditioning

3 PRINCIPLE

Specimens of known dimensions are marked, subjected to the specified treatment (washing, drying, or ironing), and the change in dimensions is measured and expressed as a percentage of the original dimension.

4 REQUIREMENTS

4.1 Dimensional Tolerance

Fabric Type	Wash Shrinkage %, max	Dry Shrinkage %, max	Iron Shrinkage %, max
Cotton Woven	5.0	3.0	2.0
Polyester Woven	2.0	1.5	1.0
Cotton Knitted	7.0	4.0	2.5
Polyester Knitted	3.0	2.0	1.0
Blended Woven	4.0	2.5	1.5
Denim	6.0	3.5	2.0

5 FASTNESS REQUIREMENTS

5.1 Color Fastness to Washing

Grade	Change in Color	Staining
5 (Excellent)	No change	No staining
4 (Good)	Slight change	Slight staining
3 (Moderate)	Moderate change	Moderate staining
2 (Fair)	Considerable change	Considerable staining

6 TEST METHODS

- Dimensional Change: IS 1758 - After 5 wash cycles at 40C
- Color Fastness to Washing: IS 11623 - Multiple Leg Method
- Color Fastness to Light: IS 106 Part 2 - Xenon Arc Lamp
- Pilling Resistance: IS 12947 - Martindale Method
- Tensile Strength: IS 1969 - Strip Method